

SPARK

Solar Particle Acceleration, Radiation and Kinetics

Resolving how solar flares accelerate particles.

SPARK is an ESA M-class mission concept designed to answer a fundamental question in astrophysics: how magnetised plasmas explosively convert stored energy into accelerated particles and heated plasma.

Mission class	ESA M-class concept	Orbit	Sun–Earth L1
Launch epoch	2041	Mission duration	4 years nominal + 3 years extended
Payload	LASSAN + HiFI	Science focus	Solar flare particle acceleration

Why solar flares?

Solar flares are the nearest explosive particle accelerators that can be resolved in space and time. They release stored magnetic energy, accelerate electrons and ions, emit intense X-rays and gamma-rays, and reshape the solar atmosphere.

What LASSAN measures

LASSAN measures the energetic particles through X-ray and gamma-ray diagnostics: bremsstrahlung from accelerated electrons, nuclear gamma-ray lines from accelerated ions, and pion-decay emission from relativistic ions.

What HiFI measures

HiFI observes the plasma response through high-cadence, high-resolution EUV imaging of flare ribbons, kernels, hot loops, waves, flows and reconnection structures.

Why SPARK is different

SPARK combines particle diagnostics and plasma imaging in a single coordinated observatory, connecting magnetic energy release to particle acceleration, energy transport and atmospheric response.

Why now?

Large-area gamma-ray detector technology, Solar Orbiter/EUI heritage and the Solar Cycle 27 opportunity make SPARK timely for a statistically meaningful flare programme.

Impact

SPARK will advance heliophysics, plasma astrophysics and space-weather science by establishing how energetic particles are produced and how flare energy is transported through the solar atmosphere.

Website resource prepared for the SPARK mission concept. Replace or extend this file as approved public material becomes available.